

SP-05: A positive-semidefinite minimizing eigenmatrix

Proposed resolution for mathematical review

Status of this manuscript: Proposed resolution, not independently verified.

Outcome claimed: Affirmative resolution claim.

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Target: [Repository entry SP-05](#), snapshot b412319.

This is a proposed argument generated in a ChatGPT conversation and prepared for mathematical review. No independent referee report or formal proof certificate accompanies it. Supporting algebraic and numerical diagnostics are not a substitute for proof review. No publication or priority claim is made.

Theorem SP-05: proposed positive-semidefinite minimizer

Claim. For any real symmetric positive definite $n \times n$ matrices A, B , the operator

$$\mathcal{L}(X) = AXB + BXA$$

has a nonzero real positive-semidefinite eigenmatrix for its smallest eigenvalue. Under column vectorization its matrix is $A \otimes B + B \otimes A$, so this implies the symmetric-minimizer assertion in SP-05.

The Frobenius inner product on complex matrices is $\langle X, Y \rangle = \text{tr}(X^*Y)$. The operator $\mathcal{L}$ is self-adjoint and positive definite; for example,

$$\langle X, AXB \rangle = \|A^{1/2}XB^{1/2}\|_F^2 > 0 \quad (X \neq 0).$$

It also commutes with transposition.

1. An inverse that preserves positive semidefiniteness

Put $C = B^{-1/2}AB^{-1/2} \succ 0$. In the equation $\mathcal{L}(X) = Y$, set

$$\tilde{X} = B^{1/2}XB^{1/2}, \quad \tilde{Y} = B^{-1/2}YB^{-1/2}.$$

Then

$$C\tilde{X} + \tilde{X}C = \tilde{Y}, \quad \tilde{X} = \int_0^\infty e^{-tC}\tilde{Y}e^{-tC} dt.$$

The integral converges since $C \succ 0$. Differentiating its integrand verifies the equation, and uniqueness follows from positive definiteness of the Sylvester operator. Thus $\Phi = \mathcal{L}^{-1}$ satisfies

$$\Phi(Y) = \int_0^\infty M_t Y M_t dt, \quad M_t = B^{-1/2}e^{-tC}B^{-1/2}.$$

Every M_t is real symmetric. Consequently Φ maps complex Hermitian positive-semidefinite matrices to positive-semidefinite matrices.

2. The top inverse eigenvalue has a real PSD eigenmatrix

Let $r = \lambda_{\max}(\Phi)$. A real eigenmatrix for r exists because Φ is real self-adjoint. Since Φ commutes with transposition, at least one nonzero symmetric or skew-symmetric part of that eigenmatrix is also an eigenmatrix for r . Call such a part W and define

$$H = \begin{cases} W, & W^T = W, \\ iW, & W^T = -W. \end{cases}$$

Then H is Hermitian and $\Phi(H) = rH$.

Write $H = H_+ - H_-$ with $H_+, H_- \succeq 0$, and $|H| = H_+ + H_-$. Self-adjointness and positivity preservation give

$$\begin{aligned} \langle |H|, \Phi(|H|) \rangle - \langle H, \Phi(H) \rangle \\ = 4 \operatorname{tr}(H_+ \Phi(H_-)) \geq 0. \end{aligned}$$

The trace is nonnegative because the trace of the product of two Hermitian positive-semidefinite matrices is nonnegative. Also $\| |H| \|_F = \| H \|_F$.

In both cases $|H|$ is real symmetric and positive semidefinite. For a real symmetric W this follows by real spectral calculus. For real skew-symmetric W ,

$$|H| = \sqrt{H^2} = \sqrt{-W^2},$$

the real positive-semidefinite square root of the real positive-semidefinite matrix $-W^2$.

The Rayleigh quotient of $|H|$ for Φ is at least r . By the variational characterization of the largest eigenvalue it is also at most r , so equality holds and $|H|$ is an eigenmatrix for r . It is nonzero because $H \neq 0$. Therefore

$$\mathcal{L}(|H|) = r^{-1}|H|, \quad r^{-1} = \lambda_{\min}(\mathcal{L}),$$

which proves the claimed stronger result.

3. The displayed symmetric/skew inequality

Let $T \operatorname{vec}(X) = \operatorname{vec}(X^T)$. Then

$$T(A \otimes B)T = B \otimes A.$$

For a vector v in either the $+1$ or -1 eigenspace of T ,

$$v^T(A \otimes B + B \otimes A)v = 2v^T(A \otimes B)v.$$

The two eigenspaces are orthogonal and invariant for the sum. Since its global minimum is attained in the symmetric sector, the minimum Rayleigh quotient of $A \otimes B$ in that sector is at most the minimum in the skew-symmetric sector. This is the precise inequality in the repository entry. $\square$

Scope and review notes

The proof uses arbitrary real symmetric positive definite A and B , without a commutativity or rank restriction. It concerns the smallest eigenvalue of $A \otimes B + B \otimes A$, not a maximum-eigenvalue problem or an indefinite-input generalization.

Review should check the congruence reduction for the inverse, use of positivity on the complex Hermitian cone, and the argument that the modulus of iW is real when W is real skew-symmetric. The proof does not assume a simple minimum eigenvalue.

The packaging pass checked the repository statement and contribution rules on 11 September 2026. It did not conduct a new exhaustive literature search or establish novelty.

Source references

Alex Townsend, *Open Problems in Numerical Linear Algebra* (2026), [entry SP-05](#), repository snapshot b412319, accessed 11 September 2026.

N. Kalantarova and L. Tunçel, *On the spectral structure of Jordan-Kronecker products of symmetric and skew-symmetric matrices*, *Linear Algebra and its Applications* 608 (2021), 343–362, [arXiv:1805.09737v3](#), Conjecture 1, equation (7), p. 12. These are the original-source locators recorded by the catalog.